

WANG MODEL 702 PLOTTING OUTPUT WRITER

The Model 702 Plotting Output Writer adds digital plotting (X, Y) and alpha/numeric output writing capabilities to Wang's Model 700B or 720B Programmable Calculator. A true dual-purpose peripheral, the Model 702 can provide any number of alpha/numeric printout and plot combinations, either by program control or manually. Faster and more accurate than an analog plotter, the 702 can produce a wide range of variable spacing plots. A variety of symbols can be used to differentiate between various functions or for superimposing plotting curves. Because plots and printouts can be combined on the same paper, the need is eliminated for time-consuming correlations of plots and printouts from various printout media. Applications for the Model 702 are found in almost every engineering and scientific discipline where the presentation of data in graphic form would be desirable.

The basic machine is an IBM Selectric® typewriter which has been modified by replacing the forward carriage feed and tabulation mechanism with a carriage-drive stepping motor for the X axis and replacing the platen drive by a second stepping motor for the Y axis. Both motors have a stepping rate of 400 steps/second in increments as small as 0.01 inches. The return mechanism enables the carriage to return at a rate of 17 inches per second.

Full capability of the manual keyboard is retained except for tabulation control. This advantage gives the user an additional typewriter during non-plotting and non-calculating periods. Programmable functions include space, backspace, index, reverse index, power on-off, shift down, shift up, and return. The reverse index key (new in design and not available on the Model 701



Output Writer) allows a reverse line feed operation. Space and index functions are conventional.

(X, Y) PLOTTING CHARACTERISTICS

The Model 702 has a four quadrant plotting capability. The origin point can be located anywhere within the parameters of the plotting area (12.25 inches on the X-axis, Y-axis is limited only by the length of the continuous form which is used).

Stepping Rate (X or Y-axis)	400 steps/second (approx. 4 inches/second)
Stepping Increment ($\Delta X, \Delta Y$)	0.01 inch/step
Plotting Time (minimum)	0.1 seconds between plot points
Plotting Accuracy	± 0.010 inches or $\pm .003$ /inch
Dynamic Accuracy	Deviation from straight lines between two data points is less than ± 0.03 inches for data points up to 10 inches apart.
Pin Feed Platen	Vertical spacing between pins is 1/2 inch. Horizontal spacing between pins is 13 1/8 inches.

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PROGRAMMING CHARACTERISTICS

Ease and range of programming the Model 702 through the Model 700B calculator provides a convenient and practical method for obtaining many previously impractical plot/print combinations. The following operations are indicative of the output flexibility a user has at his disposal with the Model 702.

Numeric Output with Decimal Point — This operation is controlled by a 2-step command. The **WRITE** key followed by a 4-digit code number will produce a printout of the X register in the desired format. Up to nine characters to either the left or right of the decimal point can be obtained.

Numeric Output with Modified Scientific Notation — This operation is controlled by a 2-step command. The **WRITE** key followed by the code 00 15 will produce a printout of the entire X register in modified scientific notation, regardless of the possible floating decimal point condition of the X register.

Space Function — Any number of spaces from 1 to 15 may be automatically programmed by a 2-step

command. The **WRITE** key followed by the code 15 XX will produce XX spaces.

Alphabetic Output — Alphabetic printouts such as titles can be printed under program control by using the **WRITE ALPHA** command. Each succeeding code number after the initial command will produce an alpha character printout. This mode is terminated by an **END ALPHA** command.

X, Y Plot — Like the alpha mode, an X, Y plot can be programmed by starting with a **WRITE ALPHA** command. Each succeeding code number will produce a character print at an X-Y distance from the previous plot point. The X-Y distance is defined by the integer portions of the X and Y registers, respectively. The series of plot points is terminated by using the **END ALPHA** command. The differentiation between the print and plot mode is made by the code number. A convenient function, called **PLOT**, enables the positioning of the plotter at any X-Y position without actually printing a plot symbol. Since both the alphabetic print and the plot print functions use the same **WRITE ALPHA** command, a user can very conveniently intermix alpha/numeric printout and plot instructions with one-step programming.

ALPHA/NUMERIC OUTPUT WRITER CHARACTERISTICS

A total of 88 characters (44 lower case and 44 upper case) is available at the keyboard of the Model 702.

Paper Width	13 5/8 inches
Width of Printing Area	12 1/4
Character/Inch	10
Lines/Inch	Approx. 7
Line Pitch	.14 inch
Type Face	IBM Prestige Elite 72 (code 012)
Maximum Printing Speed	Approx. 15.5 characters/second
Space and Back Space Speed	30 milliseconds
Index Speed (forward and reverse)	50 milliseconds
Carriage Return Speed	0.15 seconds plus 60 milliseconds/inch
Weight	Net 96 lbs, gross 140 lbs.
Connecting Cables	12 ft. output connector cable & power supply cord
Power Requirements	115 VAC $\pm$ 10%, 60 Hz, 200 W, 3 Amp. (230 VAC $\pm$ 10%, 50 Hz, 200 W, 1 1/2 Amp. on special order)

WARRANTY

Full warranty of 90 days for all parts and labor. Optional maintenance contracts are available.